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Adaptive fuzzy PI controller for permanent magnet synchronous motor drive based on predictive functional control

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Abstract

The high-performance speed control is hard to achieve for the permanent magnet synchronous motor (PMSM) drive system under uncertain disturbances, input saturation, and system delay. To address this challenge, a novel adaptive fuzzy proportional-integral (AFPI) control scheme is investigated in this paper. An anti-saturation PI (ASPI) controller is proposed serving as the main control unit that drives PMSM to track the desired profiles, which is optimized online by a designed adaptive fuzzy tuner. Specifically, together with the parameter update constraints, an improved anti-integral saturation mechanism is suggested for the ASPI controller. Meanwhile, the adaptive fuzzy tuner with the self-tuning input domain enhances the optimization ability to effectively tackle the system uncertainties. Then, a predictive functional control method is incorporated into the tuner to mitigate the control lag caused by error-driven characteristics, which ensures the dynamic tracing control performance in the presence of system delay. The stability and the H_∞ robustness of the AFPI controller are guaranteed based on D-decomposition theory. Simulations and real-time experiments are performed, and the results illustrate that the designed scheme has superior performance in terms of dynamic tracking control and anti-disturbance capabilities as compared to conventional methods.

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1. Introduction

Benefitting from the praiseworthy advantages of simple structure, clear functions, and easy implementation, proportional-integral (PI) controllers with different forms have been widely

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recognized in industrial fields. Typical applications include industrial robots or machine tools actuated by the PMSM drive systems. However, PMSM drive systems contain various nonlinear factors, which need to be considered for the controller design to fully exploit the potential of the equipment [1–3]. For instance, mechanically introduced torque disturbances (e.g., nonlinear friction) may cause dynamic performance degradation in speed regulation. The uncertainty of electrical parameters will directly affect the current tracking performance. Besides, the system delay is also a nonlinear factor that cannot be ignored for a high-performance PMSM drive system. To achieve enhanced performance concerning dynamic tracking and efficiency, the PI controllers should retain their control capacities under time-varying operating conditions [4,5].

Lots of efforts have been paid to derive adjustment strategies for PI controllers, such as heuristic optimization algorithms [6–9], iterative learning control [10,11], model-based control strategies [12–14], fuzzy rule-based tuning [15,16], and other advanced control strategies. As one of the most promising adaptive technologies, fuzzy PI control can effectively utilize the known empirical knowledge to adaptively adjust the control parameters to handle the unknown disturbances (e.g., external interferences and model uncertainties). Its advantages and effectiveness to handle the systems subject to nonlinearities and uncertainties have been demonstrated in different applications [17–20]. Note that the PI controller with fuzzy online learning ability provides a protentional way to enhance the closed-loop robustness of the PMSM drive system. Although fuzzy PI control has been employed to adjust the current or speed in some reported works [21–23], there are still some challenges that need to be further explored when extending to accurate control of the PMSM system.

1) The first challenge lies in the derivation of suitable fuzzy rules since they will directly determine the dynamic performance of the resultant system. For most nonlinear industrial applications, their complex dynamic characteristics make establishing detailed and exact fuzzy rules time-consuming or even unattainable. Given this context, PMSM drive systems often face the problem of lacking sufficient and suitable fuzzy rules to mitigate the effects of unknown disturbances. Meanwhile, the fuzzy PI controllers based only on conventional control laws are impractical to achieve high-precision control of the objective [24,25]. Some literature focuses on enhancing the performance of fuzzy controllers by optimizing the key parameters. For example, the membership functions are optimized online by a Kalman filter [26], or the fuzzy rules are tuned online via rule weight [27], and the input/output scaling factors are determined by the particle swarm optimization algorithm [28] or the dynamic just-in-time learning algorithm [29]. Although the existing methods can improve fuzzy controllability, they are still not effectively applied to a practical PMSM drive system. On one hand, the process of optimizing complex parameters is time-consuming for the motion control system that pursues high response. On the other hand, the above-mentioned fuzzy control methods lack practical theoretical analysis to ensure the stability and robustness of a resultant closed-loop system under the external disturbances and parametric uncertainties [30,31]. Therefore, it is still of great significance to explore effective strategies to design fuzzy PI controllers for the PMSM drive system.

2) Another challenge is how to address the system delay and control lag issue when deriving fuzzy PI control solutions. In the PMSM system, there are inevitable time delays caused by the sensors, filters, and electronic signal conversion. Meanwhile, there are data processing delays in the process of generating control parameter adjustments based on the fuzzy tuner. These delays can cause the update of parameters to lag behind the change of the system state

Challenges
of current
strategies

1. Fuzzy Model Predictive Control → BDK fuzzy dc xay dung dua tren mo hinh du bao (MPC).
2. Fuzzy control using predicted output as input signal → BDK fuzzy su dung tin hieu input la gia tri output cua MPC

due to the error-driven characteristics of the fuzzy tuner. In this sense, given the adaptive ability, the existing fuzzy PI control strategies are hard to effectively handle the nonlinear disturbances while improving the dynamic response. It is noted that model predictive control (MPC) enables the system to forecast future operating conditions and then make control decisions accordingly, which is effective to resolve system delays or control lag [32–34]. Therefore, some control strategies combining fuzzy control and MPC are proposed to combine the advantages of both schemes. These strategies can be generally divided into two categories: fuzzy model predictive control [35–37] and fuzzy control using predicted output as the input signal [38,39]. The former directly uses the predicted output as the driving signal of the system, which requires a high-precision model to ensure the desired control performance. In comparison, the latter can effectively utilize the control robustness to reduce the demand of the controlled system for prediction accuracy. The predictive process based on MPC will lead to a complex control structure for time-varying dynamic systems such as the PMSM servo drive [40]. As a practical improvement of MPC, predictive functional control (PFC) has the advantage to handle uncertainties and system delays with reduced computation [41,42]. In this sense, a reasonable introduction of the PFC will provide a novel insight to overcome the shortcomings of fuzzy PI controllers. Unlike the existing fuzzy predictive control or the MPC, the fuzzy PI control based on PFC can obtain effective processing delay ability while reducing the dependence on model accuracy. This scheme can be better applied to systems such as PMSM drive systems that suffer from the accurate identification of dynamic models.

Conclusion

Given the above challenges, a novel adaptive fuzzy PI (AFPI) controller is designed for the PMSM drive system subject to unknown disturbances, input saturation, and control lag. The proposed AFPI control scheme consists of an improved PFC module, an adaptive fuzzy tuner, and a novel anti-saturation PI (ASPI) controller. Meanwhile, the stability and H_∞ robustness of the resultant system under AFPI control are guaranteed. In summary, this paper has the following contributions: (1) With the real-time self-tuning input domain, an adaptive fuzzy tuner is proposed to ensure that the nonlinear factors of the PMSM drive system can be effectively handled with common fuzzy rules. Compared with [26–29], it has enhanced practicality and robustness while improving parameter tuning capability for unknown disturbances. (2) To address the system delay and control lag, the improved PFC method is presented to provide the future system state information for the fuzzy tuner, therefore pre-optimizing the control parameters. Unlike [34–36], it achieves effective predictive control capabilities under a simplified model by utilizing the robustness of fuzzy control. (3) Differing from the existing fuzzy PI controllers (e.g., [17–20]), a novel ASPI controller is presented for addressing the input saturation issue based on the predictive information. This controller specifies update constraints for the control parameter while designing an anti-integral saturation strategy, which ensures that the control output can jump out of the saturation zone in time. (4) New sufficient conditions with respect to stability and robustness of the AFPI control system are derived with the aid of the D-decomposition theory, which guarantees the global asymptotic convergence and H_∞ anti-interference performance.

The remainder of this paper is constructed as follows. Section 2 provides the problem formulation. Then, the proposed AFPI control scheme is presented in Section 3 while Section 4 offers stability and H_∞ robustness analysis. In Section 5, simulations and experiments are performed to illustrate the effectiveness and superiorities of the proposed strategy. Finally, the conclusions are drawn in Section 6.

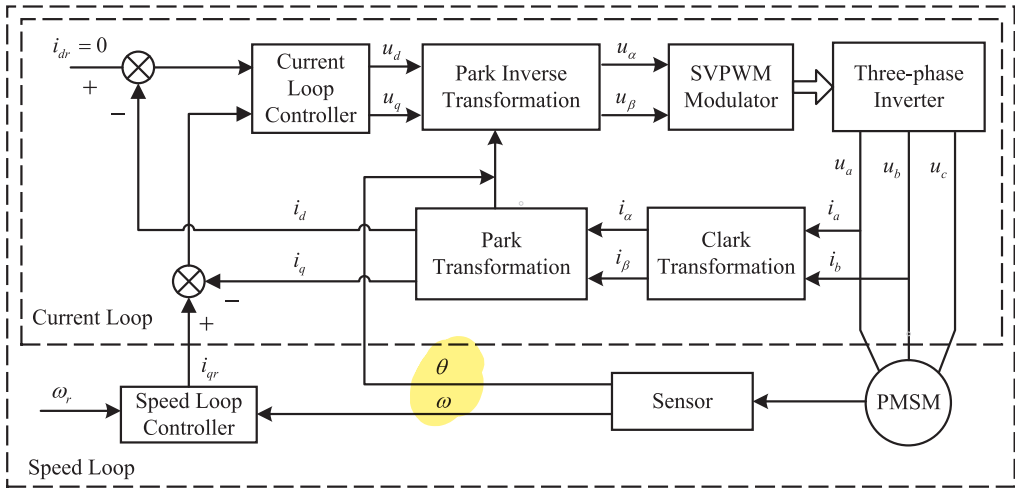


Fig. 1. PMSM speed control system.

2. Problem formulation

2.1. Main nonlinear factors in the PMSM system

In d - q axis coordinates, the dynamics of a PMSM can be described as follows

$$\dot{i}_d = (V_d - R_s i_d)/L_d + n_p L_q \omega i_q / L_d, \quad (1)$$

$$\dot{i}_q = (V_q - R_s i_q)/L_q - n_p \omega (L_d i_d + \psi_f)/L_q, \quad (2)$$

$$\dot{\omega} = (3n_p(\psi_f i_q + (L_q - L_d)i_q i_d)/2 - B_f \omega - T_l)/J_m, \quad (3)$$

where i_d and i_q are the d - q axis currents, V_d and V_q are the d - q axis voltages, ω is the rotor angular speed, T_l is the lumped torque disturbances, L_d and L_q represent the d - q axis stator inductances, n_p , R_s , J_m , B_f , and ψ_f represent the number of magnetic pole pairs, stator resistance, rotor inertia, viscous friction coefficient, and rotor flux linkage, respectively.

Fig. 1 shows the PMSM speed control system based on the field-oriented control, adopting a cascade control structure that consists of a current loop and a speed loop. In practice, the different control loops of the PMSM drive system all suffer various nonlinear impacts including internal nonlinear coupling, parametric perturbations, external disturbances, and system delay.

1) Current loop.

In the current loop, the d -axis and q -axis reference currents are 0 and the output of the speed controller, respectively. Under the regulation of the current controller, they are converted to the voltage command of the servo drive. At the same time, the voltage

command is converted into PMSM's real-time three-phase control voltage with the assistance of Clark, Park, and Park inverse transformations, SVPWM modulator, three-phase inverter, and other modules. In the voltage Eqs. (1) and (2), there exist cross-coupling terms between the d and q axes, which can be decoupled by the following formulation

$$\begin{cases} u_{df} = n_p L_q \omega i_q \\ u_{qf} = n_p L_d \omega i_d + n_p \psi_f \omega \end{cases} \quad (4)$$

Then, the d - q axis currents can be controlled independently, i.e.,

$$\begin{cases} u_d = (R_s + L_d s) i_d \\ u_q = (R_s + L_q s) i_q \end{cases} \quad (5)$$

where s denotes the Laplace variable, u_d and u_q are subject to $V_d = u_d - u_{df}$ and $V_q = u_q + u_{qf}$, respectively.

Due to physical constraints, the current loop unavoidably suffers parametric uncertainties and input saturation issues:

- 1) The inductance (L_d, L_q) variations caused by the magnetic saturation of the motor.
- 2) The stator resistance (R_s) variations with temperature.
- 3) The allowable currents and voltages with clear upper bounds.

II) Speed loop

Under the field-oriented control framework, the dynamic (3) related to the speed can be reorganized as follows

$$\dot{\omega} = (K_f i_q - B_f \omega - T_l) / J_m, \quad (6)$$

where $K_f = 3n_p \psi_f / 2$ denotes the torque coefficient.

The disturbance torque T_l mainly includes load torque, friction torque, cogging torque, etc. These nonlinear disturbances impose difficulties to design motion controllers that can achieve high-speed and high-precision performance. Using the Stribeck model, the disturbance torque is determined by

$$T_l = f(\omega) = \begin{cases} F_e, & \text{if } \omega = 0 \text{ and } |F_e| < F_s; \\ F_s \text{sign}(F_e), & \text{if } \omega = 0 \text{ and } |F_e| \geq F_s; \\ \left[F_c + (F_s - F_c) e^{-(\omega/\nu)^2} \right] \text{sign}(\omega), & \text{if } \omega \neq 0; \end{cases} \quad (7)$$

where F_e , F_s , and F_c denote the applied external load torque, the maximum static friction torque, and the Coulomb friction torque, respectively, and ν is the Stribeck velocity.

Generally, these nonlinear torques are uncertain and difficult to be accurately identified. On the other hand, they have an upper bound $T_{\max}$, i.e., $f(\omega) \leq T_{\max}$.

2.2. Fuzzy PI control system

To tackle the above challenges, the fuzzy PI control can be employed to regulate the current or speed, and its generic model can be represented as

$$u(t) = \hat{k}_p(t)(e(t) + \hat{k}_i(t) \int_0^t e(t)), \quad (8)$$

where $\hat{k}_p(t)$ and $\hat{k}_i(t)$ are the respective proportional and integral gains, which are updated based on the system state, implying

$$\begin{cases} \hat{k}_p(t) = k_{p0} + y_{\Delta k_p}(e(t), \Delta e(t)) \\ \hat{k}_i(t) = k_{i0} + y_{\Delta k_i}(e(t), \Delta e(t)) \end{cases}, \quad (9)$$

where k_{p0} and k_{i0} are the corresponding initial PI parameters, $y_{\Delta k_p}(\cdot)$ and $y_{\Delta k_i}(\cdot)$ are the adjustment increments generated by the tuner.

Benefitting from the gain self-tuning nature, the fuzzy PI controller can make timely compensation for the system state changes caused by uncertain disturbance. Thus, the fuzzy PI-controlled current loop exhibits robustness to inductance and resistance variation [21,22]. Meanwhile, the fuzzy PI speed controller can effectively alleviate the effects of the torque disturbances, where $e(t) = \omega_r(t) - \omega(t)$ with $\omega_r(t)$ being the motor command speed and $u(t)$ is the q -axis command current, i.e.,

$$i_{qr}(t) = \hat{k}_p(t)(e(t) + \hat{k}_i(t) \int_0^t e(t) dt). \quad (10)$$

Remark 1. In the real-world PMSM system, the current loop achieves a wider bandwidth than the speed loop to ensure the swiftness of the response to the control commands or parameter variations. In the speed tracking design, it is usually regarded as a unit proportional module, that is, it is assumed that the actual current accurately tracks the command current [14,15,43]. And the constructed speed control model can be described as

$$\omega(s) = K_f i_q(s) / (J_m s + B_f). \quad (11)$$

Remark 2. The above model ignores the delay effects of data processing and electronic signal conversion. And, for a practical PMSM servo system, the network feature of data transmission also leads to the inevitable delay issue. Considering the torque disturbances, input saturation, and system delays, the transfer function of the PMSM speed control system (1) is yielded as

$$\omega(s) = (K_f \text{sat}(i_q(s)) - \tilde{T}_l(s)) e^{-\tau s} / (J_m s + B_f). \quad (12)$$

where $\text{sat}(\cdot)$ is the nonlinear saturation function, τ is the lumped time delay, $\tilde{T}_l$ is the lumped uncertainty perturbation subject to $\tilde{T}_l \leq T_{\max}$.

The purpose of this paper is to establish a speed-control framework to enable that PMSM (12) can quickly and accurately track the desired profiles in the presence of nonlinear factors. Given this context, it can be formulated as the AFPI controller design problem, in which the following issues are investigated:

- 1) Although the traditional fuzzy PI controllers with sufficiently detailed empirical rules can effectively deal with various nonlinear factors, they are accompanied by a heavy system calculation burden. Besides, it is a difficult and time-consuming task to design sufficiently fuzzy rules for complex nonlinear systems.
- 2) The error-driven characteristics of the fuzzy tuner will inevitably cause the control output to lag behind the system state changes. The existence of system delays exacerbates this problem.
- 3) Most existing fuzzy PI control methods lack practical theoretical analysis for the stability and robustness of a resultant closed-loop system under uncertain disturbances.

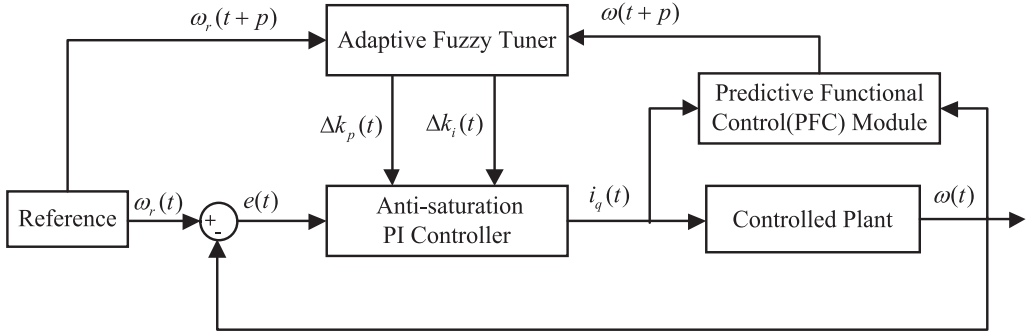


Fig. 2. The proposed AFPI control architecture.

3. The AFPI controller

As shown in Fig. 2, the AFPI control scheme is composed of a PFC module, an adaptive fuzzy tuner, and a novel ASPI controller to guarantee satisfactory tracking performance, as indicated by the following results.

3.1. The improved PFC module

The PFC module is designed to predict future outputs, which are used as the inputs of the fuzzy tuner. To formulate the predictive model, the controlled system can be described by an Auto-Regressive Xogenous (ARX) model as follows:

$$\omega(t) = (B(z^{-1})/A(z^{-1}))i_q(t), \quad (13)$$

with

$$A(z^{-1}) = 1 + a_1z^{-1} + a_2z^{-2} + \dots + a_i z^{-i} + \dots + a_{n_a} z^{-n_a}, \quad (14)$$

$$B(z^{-1}) = b_1z^{-1} + b_2z^{-2} + \dots + b_i z^{-i} + \dots + b_{n_b} z^{-n_b}, \quad (15)$$

where z^{-i} is the i order delay operating, a_i and b_i are the identified model parameters, n_a and n_b are the related orders of the model.

Referring to the ARX model, the controlled system transfer function Eq. (12) can be approximated by

$$\omega(t) = a\omega(t-1) + bi_q(t-1-\tau), \quad (16)$$

where τ is the lumped time delay, a and b are the coefficients of the model, which are identified offline by the recursive least square identification algorithm and just-in-time learning algorithm [14] in this paper.

Utilizing the Eq. (16) yields the following predictive model without time delay

$$\omega(t) = a\omega(t-1) + bi_q(t-1). \quad (17)$$

Since the unknown external disturbances and time delay are ignored, the above reference models inevitably have modeling errors. To ensure the prediction accuracy, this

section not only derives the prediction output based on the model but also proposes an error correction mechanism.

When set-points values makes step changes, the controlled plant input (i_q) will keep constant in the closed-loop response time of the PMSM drive system. Based on the PFC theory [44], it is assumed that within a limited prediction range (p), the input will remain constant in subsequent steps, i.e., $i_q(t+i) = i_q(t)$, $i = 1, 2, 3, \dots, p$. In this regards the complexity of the predictive model can be reduced. Meanwhile, the maximum prediction step can be defined as

$$p_{\max} = T_{\text{speed}}/T_{\text{current}}, \quad (18)$$

where T_{speed} and T_{current} denote the cycle times of the speed loop and current loop, respectively.

According to Eq. (17), the data of the t sampling instant are used to predict the next output of the system

$$\omega(t+1) = a\omega(t) + bi_q(t). \quad (19)$$

A straightforward calculation results in

$$\begin{aligned} \omega_m(t+2) &= a\omega_m(t+1) + bi_q(t+1) \\ &= a(a\omega_m(t) + bi_q(t)) + bi_q(t) \\ &= a^2\omega_m(t) + (a+1)bi_q(t) \end{aligned} \quad (20)$$

Similarly, the future output is predicted as

$$\begin{aligned} \omega_m(t+3) &= a\omega_m(t+2) + bi_q(t+2) \\ &= a^3\omega_m(t) + (a^2+a+1)bi_q(t) \\ &= a^3\omega_m(t) + ((1-a^3)/(1-a))bi_q(t) \end{aligned} \quad (21)$$

By mathematical induction, the system output of p steps ahead of the current sampling time is derived as

$$\begin{aligned} \omega_m(t+p) &= a\omega_m(t+p-1) + bi_q(t+p-1) \\ &= a^p\omega_m(t) + (a^{p-1} + \dots a^i + \dots a + 1)bi_q(t) \\ &= a^p\omega_m(t-1) + ((1-a^p)/(1-a))bi_q(t-1) \end{aligned} \quad (22)$$

Then, the error correction mechanism utilizes the previous prediction information to improve the prediction accuracy at $t+p$ time instant. Based on the error between the model-predicted value and the actual value, we compute the future error as

$$h(t+p) = \left(\sum_{z=0}^{p-1} m_z(\omega(t-z) - \omega_m(t-z)) \right) / \sum_{z=0}^{p-1} m_z, \quad (23)$$

where m_z is the weighting factor, $\omega(t-z)$ and $\omega_m(t-z)$ are the actual speed and model predicted speed at the time $t-z$ respectively.

Based on the revised error Eq. (23), the predicted output is corrected as

$$\omega_p(t+p) = \omega_m(t+p) + h(t+p). \quad (24)$$

Supposing the expected reference trajectory is given by

$$\omega_r(t+p) = \omega_s(t+p) - \delta^p(\omega_s(t) - \omega_p(t)), \delta = \exp^{-T_s/T_r}, \quad (25)$$

Table 1

Fuzzy rules table.

$(\Delta k_p, \Delta k_i)$	ΔE			
E	N	Z	P	
	N	(N, Z)	(N, Z)	(N, Z)
	Z	(N, P)	(Z, P)	(P, P)
	P	(P, Z)	(P, Z)	(P, P)

where T_s and T_r denote the sampling time and the closed-loop response time, respectively, ω_s is the set speed.

Then, we derive the predicted error of the $t + p$ time instant as

$$e(t + p) = \omega_r(t + p) - \omega_p(t + p). \quad (26)$$

The corresponding error derivation is described below

$$\Delta e(t + p) = e(t + p) - e(t + p - 1). \quad (27)$$

Remark 3. The improved PFC can predict the system's future states within limited steps. And, it can weaken the impact of modeling errors on the current results through the previous prediction information. Using the prediction errors as the input signal, the fuzzy tuner can respond promptly to the system state variations caused by the uncertain disturbances. Since the fuzzy control has a certain tolerance for tracking errors, this method reduces the requirements for the accuracy of prediction information. Given this context, the improved prediction function can be adapted to PMSM servo systems with different dynamic characteristics.

3.2. The adaptive fuzzy tuner *

The adaptive fuzzy tuner is then designed to automatically adjust the PI control parameters to match the current system state. The tracking error and its derivation are quantified from the actual control domain to the corresponding fuzzy domain. The actual domain and the fuzzy domain are expressed as $[-\hat{e}_m, \hat{e}_m]$ and $[-\hat{\mu}_m, \hat{\mu}_m]$, respectively. The quantified relation is defined as

$$\hat{\mu}_i = M(\hat{e}_i) = \text{sign}(\hat{e}_i) \min\{\hat{\mu}_m, |\rho \hat{e}_i|\},$$

Do không xet PFC

where $\text{sign}(\cdot)$ is the symbolic function, $\hat{e}_i$ denotes the value of $e(t + p)$ or $\Delta e(t + p)$, $\hat{\mu}_i$ denotes the value of the corresponding fuzzy variable E or ΔE , $\rho = \hat{\mu}_m / \hat{e}_m$ is the input scaling factor ρ_e or $\rho_{\Delta e}$.

Table 1 shows the fuzzy rules based on experts' experience for the regulation of PI controller parameters, where N, Z, and P represent negative, zero, and positive, respectively. According to Mamdani-type fuzzy reasoning, rule i ($i = 1, 2, \dots, 9$) can be expressed as 'If E is χ_1 and ΔE is χ_2 , then Δk_p is χ_3 and Δk_i is χ_4 ', where χ_n ($n = 1, 2, 3, 4$) represents N, Z, or P.

For the N, Z, and P, the membership functions are Gaussian, triangular, and Gaussian, respectively. Their shapes are shown in Fig. 3. In the process of fuzzification and defuzzification, each input and output variables have the same fuzzy domain and membership function shape.

N: Gauss
Z: Tri
P: Gauss

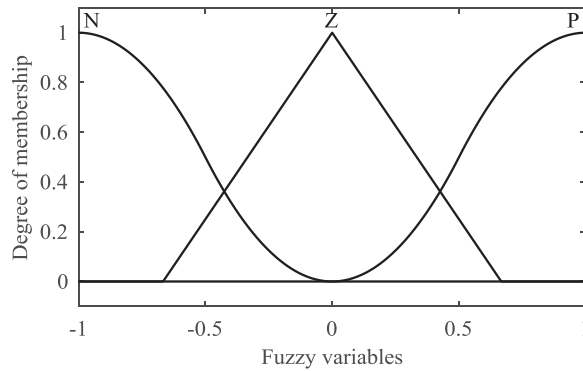


Fig. 3. Membership function.

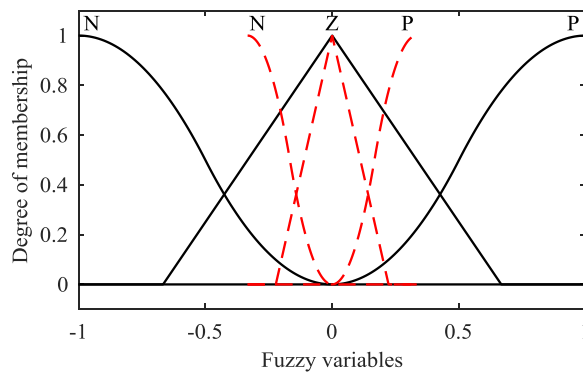


Fig. 4. Membership function with the domain shrinks (The black lines represent the initial functions, the red lines represent the functions with domain shrinks.). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

● Phan tích hạn chế của Fuzzy Tuner truyền thống/ classic

As shown in Fig. 3, near the zero points ($\hat{\mu}_i \in [-0.3, 0.3]$), the membership values of the two linguistic variables N and P are less than 0.1. This means that the traditional fuzzy tuner lacks effective rules with respect to the state changes around the steady-state value. This will seriously affect the resultant fuzzy PI controller's ability to reduce overshoot or handle sudden disturbances. To address this challenge, the contraction-expansion factor is designed for the online adjusting fuzzy input domain:

$$\lambda(\hat{\mu}_i) = (|\hat{\mu}_i|/\hat{\mu}_m)^\kappa + \varepsilon, \quad \text{He so co-dan lambda} \quad (29)$$

where $\hat{\mu}_i$ denotes the fuzzy input value in the fuzzy domain, ε is the arbitrary non-zero minimum positive value, κ is the positive value with $\kappa \in [0.5, 1)$.

The corresponding fuzzy domain is defined as follows $[-\hat{\mu}_m\lambda(\hat{\mu}_i), \hat{\mu}_m\lambda(\hat{\mu}_i)]$, where λ is the contraction-expansion factor of the domain $\hat{\mu}_m$. As shown in Fig. 4, the self-tuning input domain indirectly increases control rules around the current input. We define $g(\hat{e})$ to denote the contraction-expansion function of the input $\hat{e}$, $e(k), \Delta e(k)$

$$\begin{cases} g(\hat{\mu}_i) = \hat{\mu}_i/\lambda(\hat{\mu}_i) \\ \hat{\mu}_i = M(\hat{e}(t+p)) \end{cases}, \quad (30)$$

where $\hat{e}(t+p)$ represents the input variables $e(t+p)$ or $\Delta e(t+p)$.

Based on Eq. (30) and Mamdani-type fuzzy inference [24], the fulfillment degree of a rule is calculated by

$$\begin{cases} \tilde{f}_k(\tilde{E}_k, \Delta\tilde{E}_k) = \eta(\tilde{E}_k) \wedge \eta(\Delta\tilde{E}_k) \\ \tilde{E}_k = M(e(t+p))/g(e(t+p)) \\ \Delta\tilde{E}_k = M(\Delta e(t+p))/g(\Delta e(t+p)) \end{cases}, \quad (31)$$

where $\eta(\cdot)$ is the membership of the current inputs ($\tilde{E}_i$ or $\Delta\tilde{E}_i$), $\wedge$ is the max operator, the foot mark k represents Δk_p or Δk_i in this paper.

Employing the center of gravity algorithms, we obtain the fuzzy outputs as

$$\tilde{y}_k = \sum_{j=1}^m C_j \tilde{f}_k(e(t+p), \Delta e(t+p)) / \sum_{j=1}^m \tilde{f}_k(e(t+p), \Delta e(t+p)), \quad (32)$$

where $\tilde{y}_k$ represents the fuzzy output of the proportional gain adjustment ($\tilde{y}_{\Delta k_p}$) or the fuzzy output of the integral gain adjustment ($\tilde{y}_{\Delta k_i}$), C_j represents the discrete variable values on the fuzzy domain.

Finally, relying on the scaling factors, the fuzzy output is mapped to the actual control domain. The PI parameters adjustment values in the natural domain are calculated as follows

$$\begin{cases} \Delta k_p(t) = \alpha \cdot \tilde{y}_{\Delta k_p}(e(t+p), \Delta e(t+p)) \\ \Delta k_i(t) = \beta \cdot \tilde{y}_{\Delta k_i}(e(t+p), \Delta e(t+p)) \end{cases}, \quad (33)$$

where α and β are the output scaling factors.

Based on Eq. (33), the adaptive PI control parameters can be expressed as

$$\begin{cases} \hat{k}_p(t) = k_{p0} + \alpha \cdot \tilde{y}_{\Delta k_p}(e(t+p), \Delta e(t+p)) \\ \hat{k}_i(t) = k_{i0} + \beta \cdot \tilde{y}_{\Delta k_i}(e(t+p), \Delta e(t+p)) \end{cases} \quad (34)$$

Remark 4. By introducing the contraction-expansion factor to adjust the input domain online, the number of fuzzy rules that adjust the current system state is indirectly increased. In this way, the AFPI controller based on this scheme can process the complex nonlinear dynamics of the PMSM system based on common rules, which can further improve the robustness of the resultant system against external torque disturbances and parameter uncertainties. The pre-tuning mechanism based on predictive information ensures the response performance of the system with time delay.

3.3. The novel ASPI controller

To handle the input saturation, the nonlinear $\text{sat}(\cdot)$ function that limits the amplitude of the control signal is given by

$$i_{qr}(t) = \hat{k}_p(t)e(t) + \hat{k}_i(t) \int e(t)dt, \quad (35)$$

$$\text{sat}(i_{qr}(t)) = \begin{cases} i_{q\max}, & \text{for } i_{qr}(t) > i_{q\max}; \\ i_{qr}(t), & \text{for } i_{q\min} < i_{qr}(t) < i_{q\max}; \\ i_{q\min}, & \text{for } i_{qr}(t) < i_{q\min}; \end{cases}, \quad (36)$$

where $i_{q\max}$ and $i_{q\min}$ are upper and lower limits of q -axis input current, respectively.

Depending on the prediction information of the PFC module, the control output can be pre-evaluated by

$$i_{qr}(t+z) = \hat{k}_p(t+1)e(t+z) + \hat{k}_i(t+1) \int_0^{t+z} e(t+z)dt, \quad (37)$$

where $z = 1, 2, \dots, p$. and p denotes the prediction step.

From Eq. (35)-Eq. (37), we design the parameter update rules of the ASPI controller as

$$A(t+1) = \begin{cases} A(t) + \Delta A(t), & \text{if } i_{q\min} < i_q(t+z) < i_{q\max}; \\ A(t) + \Delta A(t), & \text{if } (i_q(t+z) > i_{q\max} \text{ and } \Delta A(t) < 0) \\ & \text{or } (i_q(t+z) < i_{q\min} \text{ and } \Delta A(t) > 0); \\ A(t), & \text{if } (i_q(t+z) > i_{q\max} \text{ and } \Delta A(t) > 0) \\ & \text{or } (i_q(t+z) < i_{q\min} \text{ and } \Delta A(t) < 0); \end{cases}, \quad (38)$$

for $\forall z \in \{1, \dots, p\}$, where A represents each control parameter, i.e., $\hat{k}_p$ or $\hat{k}_i$.

Then, we obtain the following output of the ASPI controller

$$i_{qr}(t+1) = \begin{cases} k_p(t+1)e(t), & \text{if } (i_{qr}(t+1) > i_{q\max} \text{ and } e(t) > 0) \\ & \text{or } (i_{qr}(t+1) < i_{q\min} \text{ and } e(t) < 0); \\ \hat{k}_p(t+1)e(t) + \hat{k}_i(t+1) \int e(t)dt, & \text{else;} \end{cases} \quad (39)$$

Remark 5. The improved ASPI controller provides guidelines for the output of control signals and the online update of PI parameters. Specifically, the q -axis input current is firstly pre-evaluated based on Eq. (37). Only when all the outputs from $i_{qr}(t+1)$ to $i_{qr}(t+p)$ are within the allowable interval, the control gain will be updated; otherwise, only the reverse gain will be updated to cope with the output saturation. And, the control output is defined by the integral separation principle as shown in Eq. (39). Thus, it avoids the system from sinking deeply into the nonlinear saturation region.

4. Stability and H_∞ robustness verification

4.1. Stability and H_∞ robustness analysis based on d-decomposition theory

For a PMSM speed system, the frequency response model can be represented as

$$P(jw) = N(jw)/D(jw) = R(w) + jI(w), \quad (40)$$

where $j^2 = -1$, w is the angular frequency, $N(jw)$, $D(jw)$, $R(w)$, and $I(w)$ respectively represent the numerator, denominator, real part, and imaginary part of the frequency response.

Then, the frequency domain expression of the proposed AFPI controller is

$$C_{AFPI}(jw) = \hat{k}_p(1 + \hat{k}_i/jw), \quad (41)$$

where $\hat{k}_p$ and $\hat{k}_i$ are the adaptive proportion and integral gain, respectively.

As studied in [45], the system's anti-disturbance ability can be evaluated using weighted H_∞ performance criteria. In this case, we specify the following disturbance attenuation norm for the AFPI controlled PMSM drive system

$$J_{H_\infty} = \max_{w \in [0, \infty)} |G_{yu}(jw)/jw| \leq \gamma \quad (42)$$

where γ is the desired level of bounded disturbance suppression, $1/jw$ is the weighted frequency, $G_{yu}(jw)$ is the disturbance sensitivity function that can be yielded through the combination of Eq. (40) and Eq. (41)

$$G_{yu}(jw) = P(jw)/(1 + P(jw)C_{AFPI}(jw)). \quad (43)$$

To facilitate the derivation of stability and H_∞ robustness conditions, the following definitions are given in advance.

Definition 1. The system under the adaptive controller is stable if the adaptive control parameters are bounded and the system using any fixed combination of control parameters is stable.

Definition 2. With a self-tuning controller, it is denoted that the resultant system satisfies the H_∞ disturbance attenuation index (γ), if a system using any fixed combination of control parameters meets the H_∞ disturbance attenuation index (γ).

Then, utilizing the D-decomposition theory, we derive the stability and H_∞ robustness conditions of the AFPI control system.

Theorem 1. Given a PMSM drive system, if the $\Omega_f = \Omega_s \cap \Omega_{H_\infty}$ and $K \subseteq \Omega_f$, then the proposed AFPI control determined by Eq. (34) can guarantee global asymptotic stability with anti-disturbance performance $J_{H_\infty} \leq \gamma$, where Ω_s and Ω_f denote the PI parameter domains that meet the stability constraints and H_∞ robust constraints $J_{H_\infty} \leq \gamma$, respectively, K is the set of control parameters generated by the AFPI controller and $K = \{k = [\hat{k}_p, \hat{k}_i] | k_{p0} - \alpha \leq \hat{k}_p \leq k_{p0} + \alpha, k_{i0} - \beta \leq \hat{k}_i \leq k_{i0} + \beta\}$.

Proof: According to the basic principle of Mamdani-type fuzzy control, the output is bounded. For the proposed AFPI controller, the fuzzy output domain of the adaptive fuzzy tuner is $[-1, 1]$, i.e., $y_{\Delta k_p} \in [-1, 1]$ and $y_{\Delta k_i} \in [-1, 1]$. Then, the related intervals of the adaptive PI parameters generated by Eq. (34) can be derived as $\hat{k}_p \in [k_{p0} - \alpha, k_{p0} + \alpha]$ and $\hat{k}_i \in [k_{i0} - \beta, k_{i0} + \beta]$, respectively, implying

$$K = \{k = [\hat{k}_p, \hat{k}_i] | k_{p0} - \alpha \leq \hat{k}_p \leq k_{p0} + \alpha, k_{i0} - \beta \leq \hat{k}_i \leq k_{i0} + \beta\}. \quad (44)$$

Note that, by analyzing the zero point distribution of the closed-loop characteristic equation, D-decomposition theory can determine the variation ranges of the control parameters to meet the stability and robustness requirements of the controlled system [46]. According to Eq. (40) and Eq. (41), the closed-loop characteristic equations of the PMSM speed control system can be given by

$$Q(w, \hat{k}_p, \hat{k}_i) = jwD(jw) + (\hat{k}_p jw + \hat{k}_i)N(jw), \quad (45)$$

where $w \in [0, \infty)$.

Combining the D-decomposition theory and Eq. (45), we define the stable boundary trajectories in the control parameter space as follows.

(i) Real root boundary ($\partial D_0 : w = 0$).

Substituting $w = 0$ into Eq. (45), one can obtain $\Delta(jw) = \hat{k}_i N(jw)$. Hence, the stability boundary is $\hat{k}_i = 0$ when $N(jw) \neq 0$. Otherwise, there is no boundary under $w = 0$.

(ii) Infinity root boundary ($\partial D_\infty : w = \infty$).

When $w = \infty$, the stability boundary depends on the relative order between the control system and the applied controller. In this paper, the infinity root boundary does not exist.

(iii) Complex root boundary ($\partial D_w : w \in (0, +\infty)$).

For $w \in (0, +\infty)$, the characteristic Eq. (45) is rewritten as

$$Q(w, \hat{k}_p, \hat{k}_i) = jw + (\hat{k}_p jw + \hat{k}_i)(R(w) + jI(w)). \quad (46)$$

Forcing the real and imaginary parts of Eq. (46) to zero results in the PI gains as follows

$$\begin{cases} \hat{k}_p = -R(w)/(R^2(w) + I^2(w)) \\ \hat{k}_i = -wI(w)/(R^2(w) + I^2(w)) \end{cases}. \quad (47)$$

Through the above three stable boundaries (∂D_0 , ∂D_∞ , and ∂D_w), the PI parameter stable region (Ω_s) is segmented from the entire parameter space. Moreover, the characteristic equation from the stable region has no root in the right plane of the complex plane, implying that the pole of the closed-loop system is in the left plane.

Likewise, the parameter domain that satisfies the expected disturbance attenuation index is determined through the D-decomposition theory, which is similar to the solution process of the stable domain. For the PMSM speed control system described in Eq. (42), the characteristic equation is

$$\varphi(jw, \hat{k}_p, \hat{k}_i, \gamma, \theta) = jwD(jw) + (\hat{k}_p jw + \hat{k}_i)N(jw) + \exp^{j\theta}N(jw)/\gamma, \quad (48)$$

where $\exp^{j\theta} = \cos \theta + j \sin \theta$, $\theta \in [0, 2\pi)$ and $w \in [0, \infty)$.

Then, the parameter field (Ω_{H_∞}) that satisfies the H_∞ performance index ($J_{H_\infty} \leq \gamma$) can be determined by the following three boundaries.

(i) Real root boundary ($\partial H_0 : w = 0$).

Substituting $w = 0$ into Eq. (48), then $\varphi(w, \hat{k}_p, \hat{k}_i, \lambda, \theta) = (\exp^{j\theta}/\gamma + \hat{k}_i)N(jw)$ is obtained. Hence, the robustness boundary is $\hat{k}_i = \pm 1/\gamma$ when $N(jw) \neq 0$. Otherwise, there is no boundary under $w = 0$.

(ii) Infinity root boundary ($\partial H_\infty : w = \infty$).

Similar to the stable boundary ∂D_∞ , the infinity root boundary does not exist.

(iii) Complex root boundary ($\partial H_w : w \in (0, +\infty)$).

When $w \in (0, +\infty)$, the characteristic Eq. (48) is determined by

$$\varphi(w, \hat{k}_p, \hat{k}_i, \gamma, \theta) = jw + (\hat{k}_p jw + \hat{k}_i)(R(w) + jI(w)) + (\cos \theta + j \sin \theta)(R(w) + jI(w)/\gamma). \quad (49)$$

Let the real and imaginary parts of Eq. (49) be zero, then one can obtain the following PI gains

$$\begin{cases} \hat{k}_p = -R(w)/(R^2(w) + I^2(w)) - \sin \theta/(\gamma w) \\ \hat{k}_i = -wI(w)/(R^2(w) + I^2(w)) - \cos \theta/\gamma \end{cases}. \quad (50)$$

Reviewing the design process of the Ω_s and $\Omega_{H\infty}$, it is easy to know that any control parameter (k) in the K can not only ensure the system stability but also meet the disturbance attenuation index Eq. (42), when $K \subseteq \Omega_f$ and $\Omega_f = \Omega_s \cap \Omega_{H\infty}$.

Thus, based on **Definitions 1** and **2**, it can be proved that the proposed AFPI control defines by Eq. (34) can ensure global asymptotic stability and has anti-disturbance performance $J_{H\infty} \leq \gamma$. Here completes the proof.

4.2. The design criteria of the initial parameters based on stability and H_∞ robustness conditions

In the analysis of **Section 4.1**, the parameter domains Ω_s and $\Omega_{H\infty}$ can be obtained by the frequency response of a given PMSM speed control system. Then, the parameter domain Ω_f is also determined by solving the intersection of the Ω_s and $\Omega_{H\infty}$. Consequently, the stability and H_∞ robustness of the AFPI control system mainly depend on the initial PI gains (k_{p0} , k_{i0}) and the output scaling factors (α , β) of the adaptive fuzzy tuner. The stability and H_∞ robustness conditions of the AFPI controller can be represented by the design criteria of the initial parameters. This design criterion includes the following steps:

- (i) Solve the parameter domain Ω_f using the identified frequency response of the controlled system.
- (ii) Select initial PI parameters (k_{p0} , k_{i0}) from the Ω_f based on some well-recognized rules (e.g., Ziegler-Nichols (Z-N) methods or differential evolution algorithms).
- (iii) Solve the parameters closest to the initial PI parameters on the boundary of the Ω_f , i.e., (k_p^Ω, k_i^Ω) .
- (iv) Calculate the upper bound $(\bar{\alpha}, \bar{\beta})$ of the output scaling factors as

$$\begin{cases} \bar{\alpha} = |k_p^\Omega - k_{p0}| \\ \bar{\beta} = |k_i^\Omega - k_{i0}| \end{cases}. \quad (51)$$

- (v) Determine the output scaling factors from the set $(0, \bar{\alpha}]$ and $(0, \bar{\beta}]$.

As shown in **Fig. 5**, the control parameter set K generated by AFPI is always a subset of the Ω_f under the stability and H_∞ robustness conditions. Therefore, the design criteria guarantee the closed-loop stability and H_∞ robustness of the resultant AFPI control system.

*** Remark 6.** It is worth emphasizing that the above-mentioned derivation process of stability and robustness conditions can be extended to fuzzy PI current control design

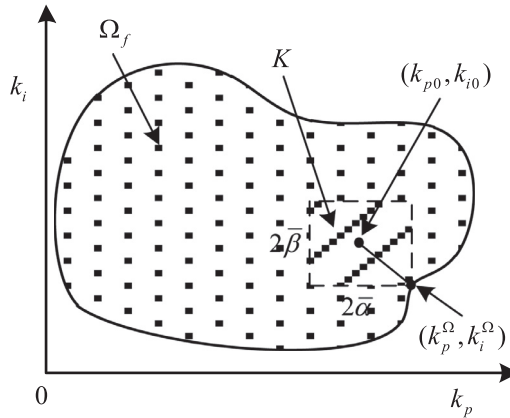


Fig. 5. Initial control parameter space and control parameter subset of the AFPI controller.

when using the frequency response model of the current loop to replace Eq. (40). Besides, the adaptive fuzzy tuner and improved ASPI controller can be directly applied to the current loop when utilizing the current command and feedback as the control input. Given this context, the proposed scheme can be appropriate to tackle the tuning of the current controllers, thereby improving the robustness of the PMSM system against internal parameter uncertainties, especially resistance and inductance variations.

5. Simulation and experiment

In this section, the proposed AFPI control method is applied in both the simulation model and the actual system to verify its practicability for the PMSM speed control application. Fig. 6 shows the control parameter adjustment process, which consists of off-line pre-tuning and online optimization to implement AFPI speed control strategies. For comparison, the traditional fixed-parameters PI control and fuzzy PI control are considered to analyze the transient response, speed tracking, and anti-disturbance capabilities.

The real-time PMSM drive system is shown in Fig. 7, which includes the following three modules:

- 1) A host computer. The host computer uses the home-developed commercial software named servo tuning platform (STP) to send the speed command and receive the feedback signal. The communication protocol adopted by the STP is a custom serial communication protocol with a cyclic redundancy check function. The maximum transmission baud rate can be set to 921,600 bit/s. In this paper, the communication cycle time is 1ms. The sampling time of the data is 0.1ms. That is, each communication will transmit 10 sets of sample data.
- 2) A servo driver. The servo driver employs STRSTM32F103 as the main processor of the control board and FPGA as the coprocessor. To obtain a higher response speed, the AFPI speed controller is implemented in the STM32. The corresponding programming tool is the IAR embedded workbench. The cycle settings of the current loop and speed loop are set to 0.05ms and 0.2ms, respectively. The driver hardware platform includes multi-channel analog and digital signal conversion and encoder circuit

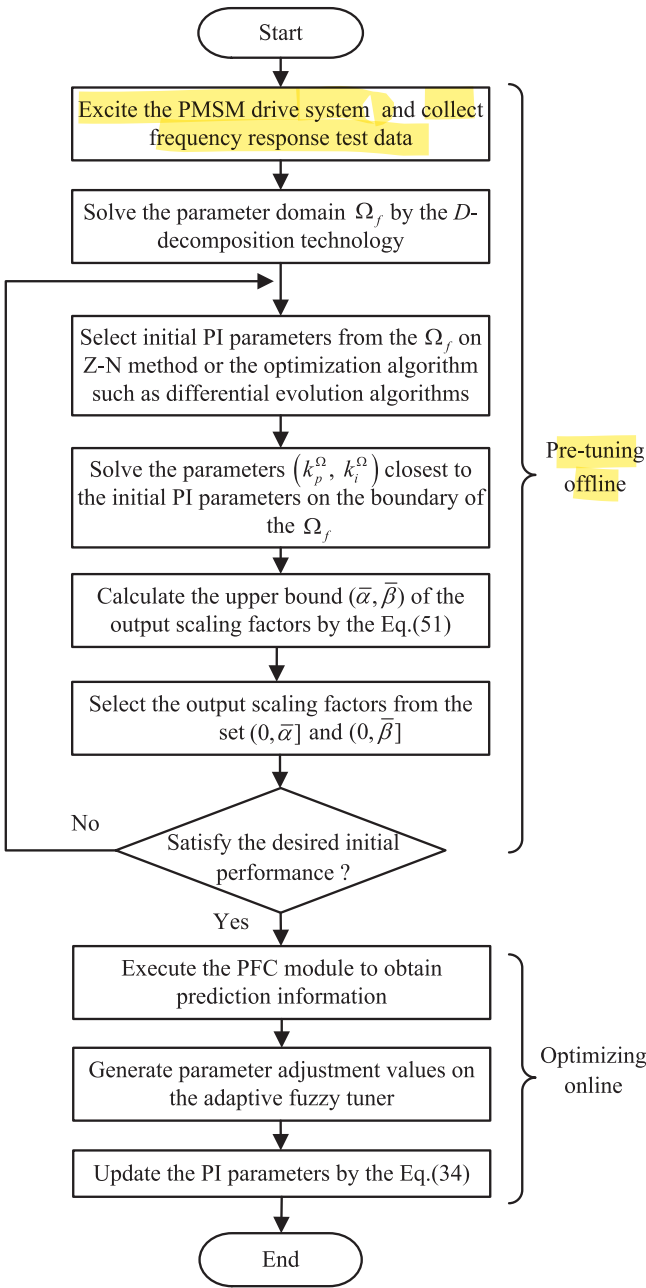


Fig. 6. Implementation procedure of the APFI control method.

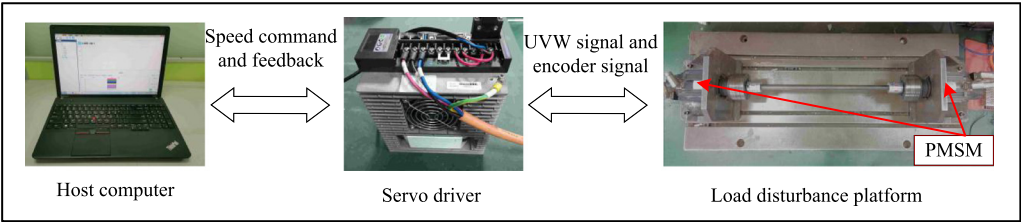


Fig. 7. Experiment platform.

Table 2
Specifications of the laboratory-made servo driver.

Characteristics	Value and Unit
Input voltage	Single/three-phase AC 220 V
Rated current	21 A
Peak current	56 A
PWM control way	Sine wave vector control
Carrier frequency	10 kHz

Table 3
Specifications of the PMSM. Tham khảo cách viết specs của PMSM

Parameters	Values	Parameters	Values
Pole pairs	5	Rated speed	2000 r/min
Power	1.3 kW	Maximum speed	3000 r/min
Rated current	6.5 A	Motor moment of inertia	$1.061 \times 10^{-3} \text{ kg}\cdot\text{m}^2$
Maximum current	20.5 A	Motor opposite electromotive force constant	$3.0814 \times 10^{-4} \text{ V/r/min}$
Rated torque	$6.4 \text{ N}\cdot\text{m}$	Phase resistance	0.6Ω
Maximum torque	$19.2 \text{ N}\cdot\text{m}$	Phase inductance	$1.88 \times 10^{-3} \text{ H}$

interfaces for signal processing. The key parameters of the PMSM servo driver are shown in Table 2.

- 3) A load disturbance platform with two PMSMs. The motor on the left is the driving component, while the other provides load disturbance. The photoelectric encoder is installed on the rotating shaft of the PMSM. It samples high-precision real-time speed signals based on the Tamagawa 23-bit absolute serial port protocol. The servo motors adopt 130ST-M0421530 PMSM of Huada New Motor Company, and its specifications are shown in Table 3.

For algorithm implementation, the prediction step size is determined by 3. The weighting factor m_z ($z = 0, 1, 2$.) is set as $1/3$. For the AFPI controller, the parameters of the adaptive fuzzy tuner are defined as $\rho_e = 5 \times 10^{-4}$, $\rho_{\Delta e} = 0.2$, $\varepsilon = 0.01$, and $\tau = 0.8$. To ensure a fair comparison, the fuzzy PI controller and the AFPI controller use the same control parameters.

According to the control structure and parameters of the real system, the corresponding simulation model is built in the MATLAB R2107a / Simulink platform.

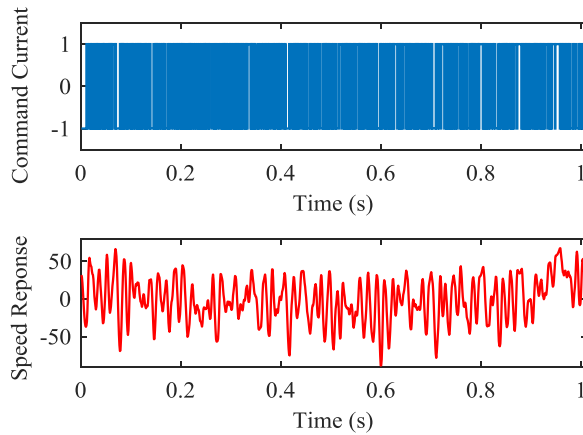


Fig. 8. Input and output signals in the identification experiment.

5.1. Simulations

In simulations, the effectiveness of the PFC module is firstly verified under the two typical input signals (i.e., step and sinusoidal speed commands). These two signals are typical signals commonly used to test system response and tracking accuracy. Then, performance evaluation experiments are performed to verify the effectiveness of the AFPI control scheme for different working conditions.

1) PFC module verification

Chu y: Khi mô phỏng trong Simulink cung cấp 1000rpm

Before performing the PFC module, the predictive function needs to be established by an offline open-loop identification experiment. The input signal is the pseudo-random binary signal. The given q -axis current inputs and corresponding outputs collected are shown in Fig. 8. Then, the initial parameters are obtained by the recursive least square algorithm and just-time learning algorithm [14]: $a = -1.003$, $b = 0.0533$.

Firstly, the step signal is selected as the command signal with an amplitude of 1000 r/min. The predicted value of the step response at time t is compared with the actual feedback value of the system running to this moment. According to Fig. 9, the maximum error between the predicted speed feedback value and the actual value occurs at the beginning, which is approximately 16 r/min. However, the maximum error only is 1.6% of the actual value (1000 r/min). It is worth mentioning that the predicted feedback value inevitably has a certain deviation compared with the actual one because of the self-tuning PI parameters. The PFC module can obtain satisfactory prediction accuracy for the step input signal.

Then, the sinusoidal signal is used as the command signal with the amplitude being 500 r/min and frequency being 5 Hz. The speed feedback values under tracking control are shown in Fig. 10. It is founded that the PFC method still has excellent prediction accuracy. In the term of the tracking error used by the AFPI method, the maximum error of the predicted result relative to the real result is 0.6334 r/min, when the actual

Step:

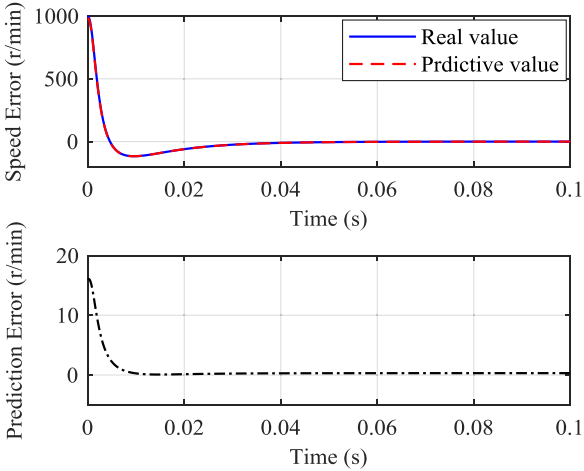


Fig. 9. Predicted speed error under the step response. (The prediction error represents the error between the predicted value and the real value.). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Sinusoidal:

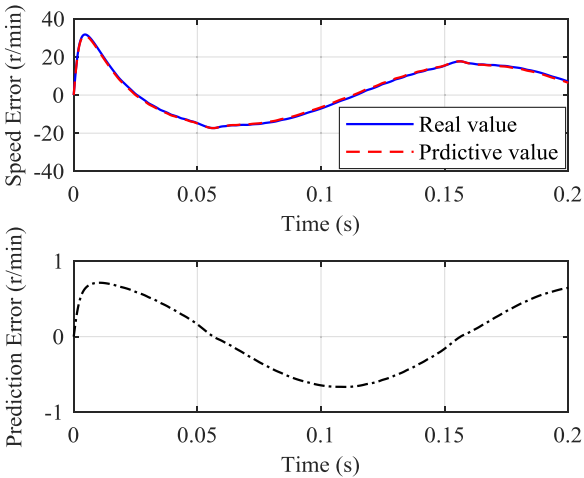


Fig. 10. Predicted speed error under sinusoidal input signal. (The prediction error represents the error between the predicted value and the real value.). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

tracking error is about 30.01 r/min. This indicates that the minimum prediction accuracy is above 97.9%. Therefore, the PFC module also satisfies the dynamical input condition.

I) The performance of the proposed method verification

As shown in Fig. 11, the stable and H_∞ region of the control gains k_p and k_i are firstly identified offline based on the design method in Section 4.1. The H_∞ disturbance attenuation index γ is 1. The data collected in the identification process comes from the input and output signals as shown in Fig. 8. According to the differential evolution

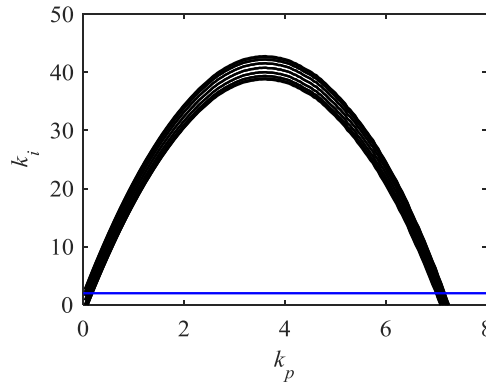


Fig. 11. Stable and H_∞ parameter region of the simulation system (Closed area enclosed by blue and black lines). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

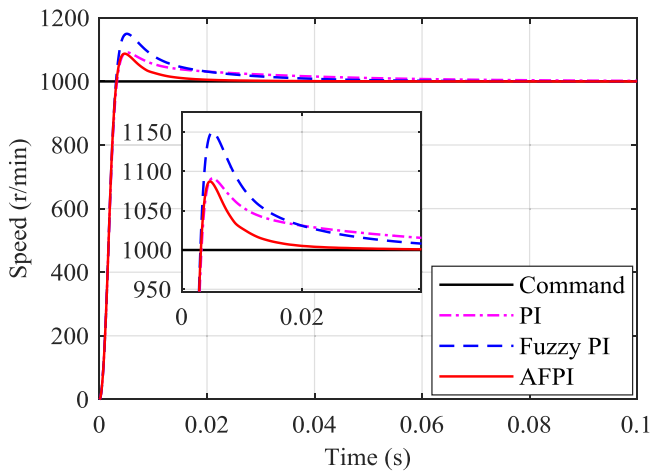


Fig. 12. Transient response of PI, fuzzy PI, AFPI in simulations.

algorithm [7], the initial parameters are defined as $k_{p0} = 2.6$, $k_{i0} = 12$. Then, the scaling factors are chosen through the design criteria of the initial parameters. The fuzzy PI controller and the AFPI controller share the same initial parameters of the fuzzy tuner, i.e., $\alpha = 1.5$, $\beta = 3$.

Fig. 12 shows the speed change curves related to transient response performance under the proposed AFPI controller, fuzzy PI controller, and PI controller. The corresponding controller outputs are given in Fig. 13. Table 4 summarizes the transient response performance indicators of three compared controllers. For the settling time index, the value 25.9ms of the fuzzy PI control system is significantly smaller than the value 32.8ms of the fixed-parameter PI control system. But, the performance advantages of the two are opposite in terms of the overshoot index. This manifests that, the fuzzy PI control system can improve the system's response performance but also may cause a decrease in relative stability due to the lack of fuzzy rules and control delay. However,

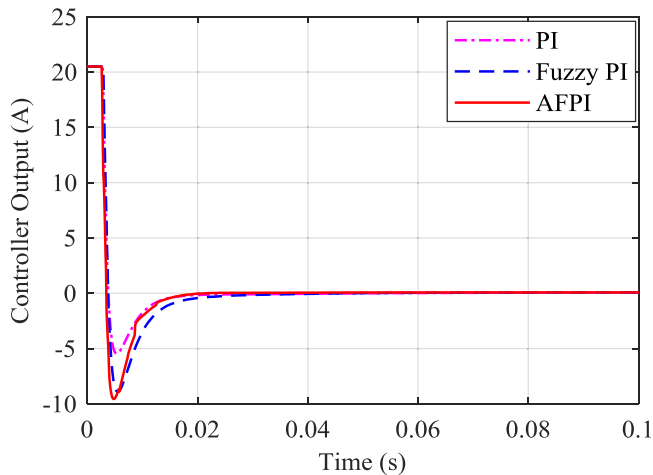


Fig. 13. Controller outputs in simulations related to the transient response performance.

Table 4
Performance indexes of PI, fuzzy PI, AFPI in simulations.

	Performance Indexes	PI	Fuzzy PI	AFPI
Transient response	Rising time (ms)	3.3	3.1	3.2
	Settling time (ms)	32.8	25.9	11.6
	Maximum overshoot (%)	9.1	15	8.7
Speed	Maximum error (r/min)	36.6	24.85	19.6
tracking	Root Mean Square Error (r/min)	26.3	18	14.2

the proposed AFPI controller performs better in terms of settling time and overshoot. The maximum overshoot of the AFPI method accounts for approximately 58% of that of the PI and about 95.6% of that of the fuzzy PI. The settling time of the AFPI method is reduced to about 1/2 of that under the fuzzy PI controller and about 1/3 of that under the PI controller. It indicates the PMSM system with the AFPI controller can have a superior transient response performance.

At the same time, an external load of 2 N•m is applied to the PMSM system at 0.1s to verify the load variation regulation capability of the proposed controller. Fig. 14 shows the speed response curves of three compared control systems when external load changes suddenly. In terms of the recovery time, the value of the PI method, the fuzzy PI, and the AFPI method are 23.7ms, 17.4ms, and 4.2ms, respectively. At the same time, it can be found from Fig. 15 that the controller output of the proposed scheme can respond to speed error fluctuations caused by the external load in time because of its excellent gain self-tuning capability. These results reveal that the proposed control scheme is better than the other two in improving the robustness of the PMSM system against uncertain disturbances.

The tracking performances of three compared controllers are illustrated in Fig. 16, where the input speed command is a sinusoidal signal with an amplitude of 1000 r/min and a frequency of 5Hz. As further observations, Figs. 17 and 18 present the corresponding controller outputs and the tracking errors, respectively. As shown in

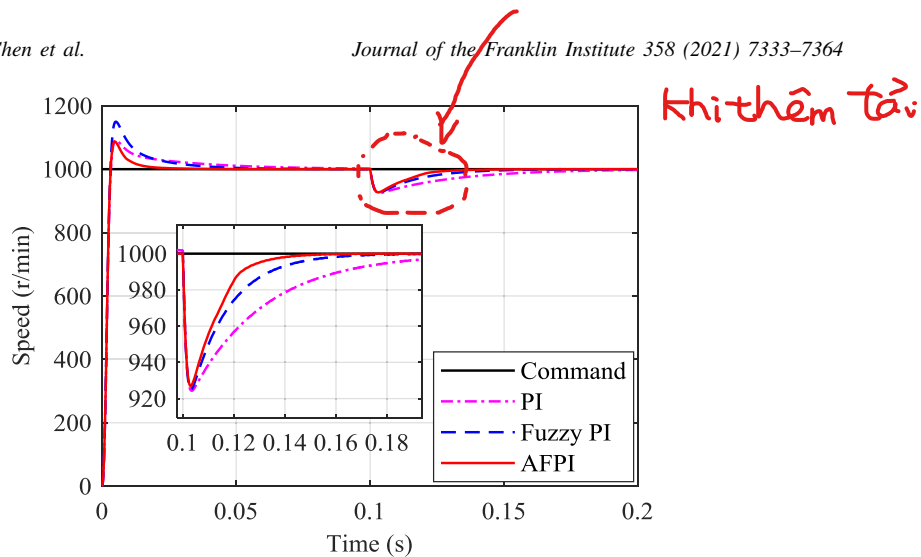


Fig. 14. Speed response curves under external disturbances in simulations.

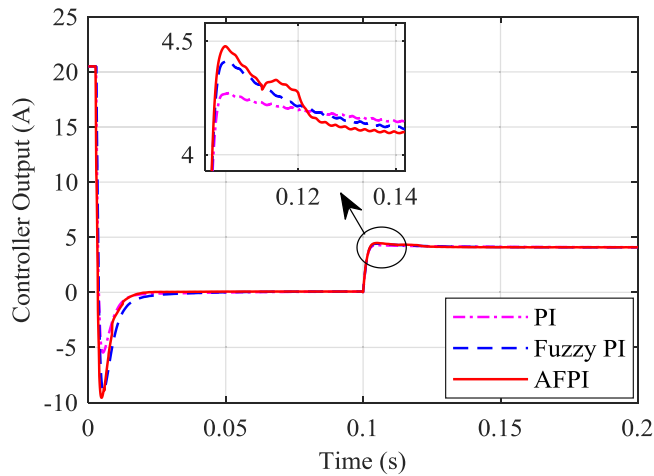


Fig. 15. Controller outputs under external disturbances in simulations.

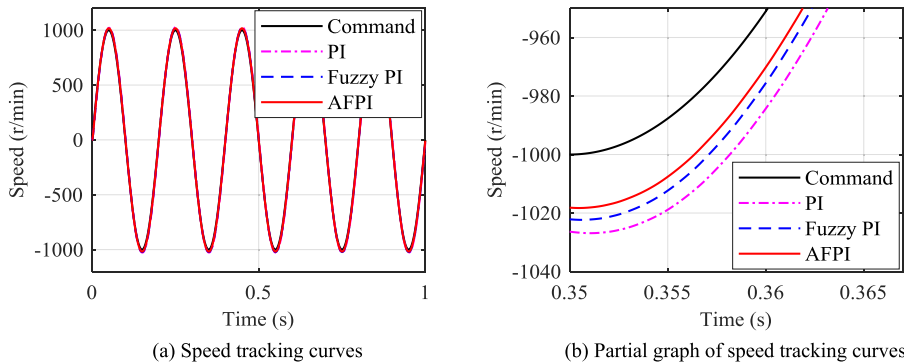


Fig. 16. Speed tracking curves of the systems using PI, fuzzy PI, AFPI.

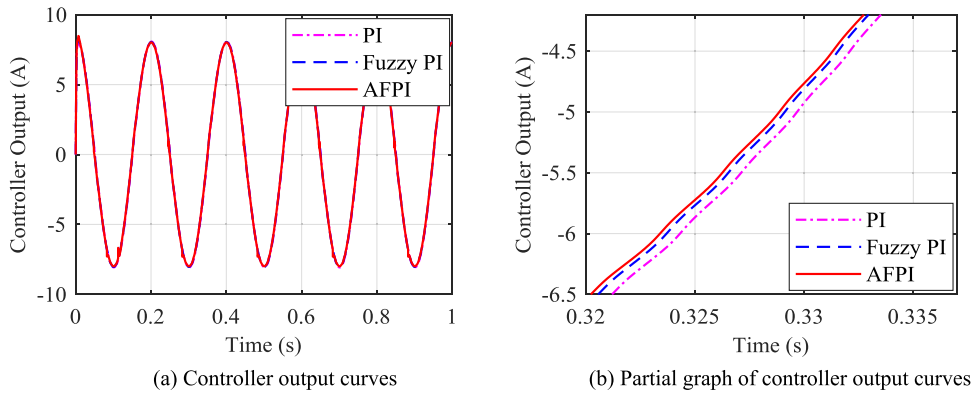


Fig. 17. Controller outputs in simulations related to speed tracking.

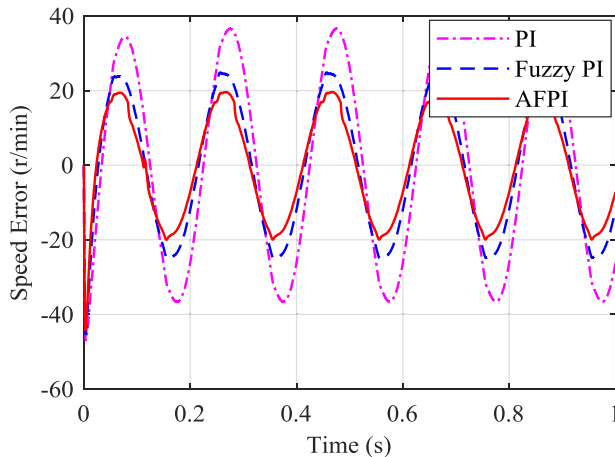


Fig. 18. Tracking error of systems using PI, fuzzy PI, AFPI.

Fig. 16, the speed tracking curve of the system using the AFPI controller is closer to the command curve. In particular, according to the results of the tracking error shown in Fig. 18, the speed tracking error of the PI control system is twice that of the AFPI control system. Corresponding error performance indexes are given in Table 4. These results show that the AFPI controller also performs well in tracking performance.

5.2. Experiments

For the considered PMSM drive system, its control parameters domain satisfying the stability and H_∞ robustness is presented in Fig. 19. The H_∞ disturbance attenuation index γ is 0.5. The identification technology is similar to the simulation process. The initial PI parameters of three controllers are set as $k_{p0} = 3.1$, $k_{i0} = 9.8$, which are given by the differential evolution algorithm. It can be seen that the initial parameters are in the stable and H_∞ region. Then, the output scaling factors are chosen as $\alpha = 1$, $\beta = 4$.

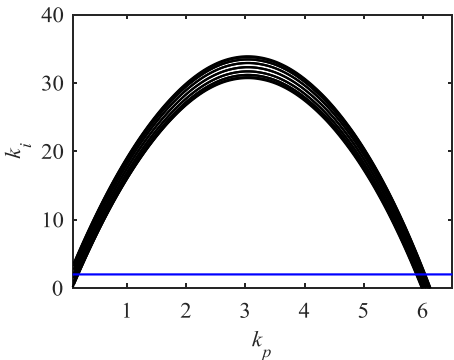


Fig. 19. Stable and H_∞ parameter region of the actual PMSM system (Closed area enclosed by blue and black lines). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

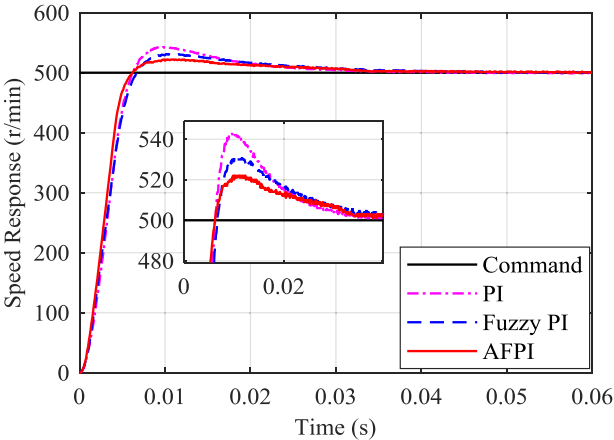


Fig. 20. System transient response when using PI, fuzzy PI, AFPI in the experiments.

Table 5
Performance indexes of the step response in the experiments.

Indexes		PI	Fuzzy PI	AFPI
Transient response	Rising time (ms)	6.3	6.6	6.2
	Settling time (ms)	24.2	24	22.5
	Maximum overshoot (%)	8.6	6.2	4
Speed tracking	Maximum error (r/min)	49.5	31	26.5
	Root Mean Square Error (r/min)	36.1	20.02	12.24

Firstly, transient response performance evaluation is executed in the PMSM servo system, where the speed input reference is set to 500 r/min. Fig. 20 demonstrates the AFPI controller remains the best dynamic responsiveness. Taking the maximum overshoot as an example, the value of the PI method, the fuzzy PI method, and the AFPI method are 8.6%, 6.2%, and 4%, respectively. The detailed transient response performance indexes can be referred to the Table 5. It cannot be ignored that the PI control strategy

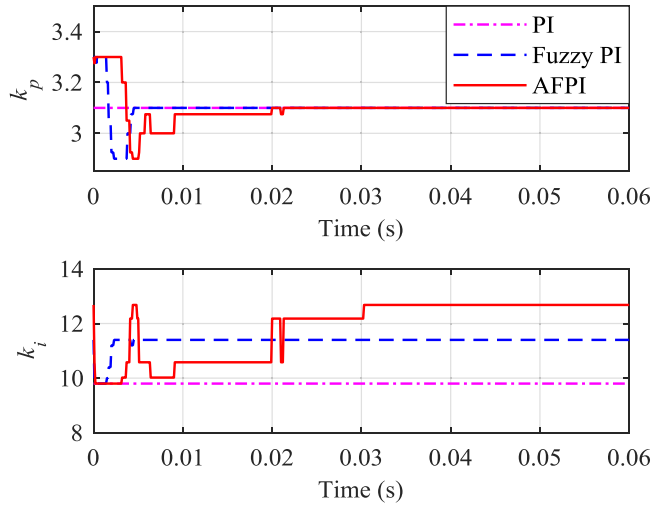


Fig. 21. Parameter (k_p , k_i) update curves for three controllers in the experiments.

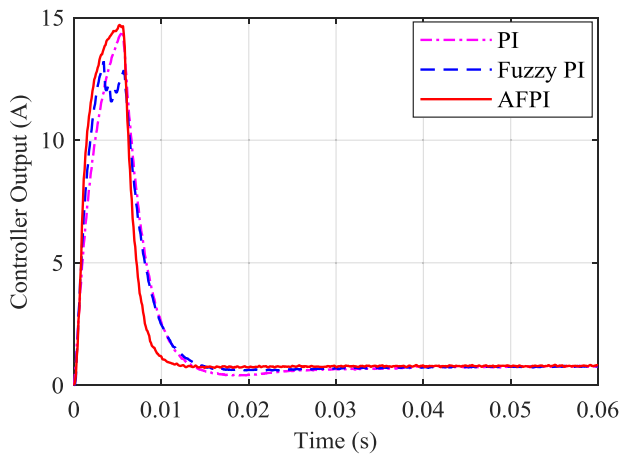


Fig. 22. Controller outputs in experiments related to the transient response performance.

outperforms the fuzzy PI control strategy in terms of rising time. The traditional fuzzy PI control can hardly balance the response performance and relative stability of the PMSM system, especially under system delay. As shown in Fig. 21, it has to reduce the proportional gain in advance to mitigate the overshoot. The proposed AFPI control strategy can solve this problem well through the improved PFC module and an adaptive fuzzy tuner. It can be seen from Figs. 21 and 22 that the control output of the proposed scheme has a better response with the help of gain auto-tuning.

Then, Fig. 23 shows the speed variation of the drive system under the external disturbance. Here, the disturbance signal is selected as an external load with amplitude 2 N·m lasting 5 speed loop cycles. It can be found that the PMSM system with fuzzy learning rate is not only least affected by the disturbance, but can recover faster. This

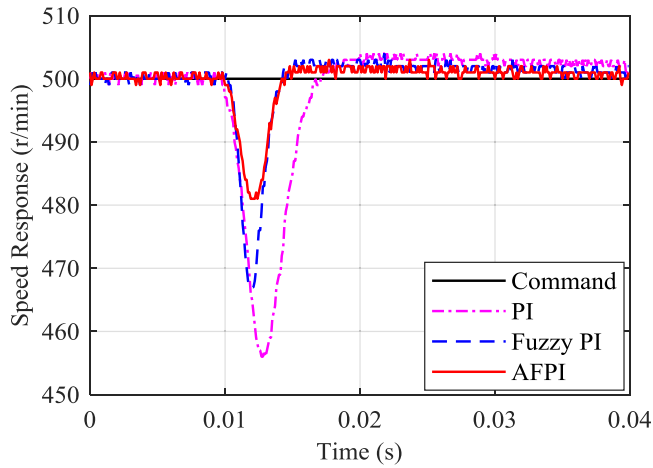


Fig. 23. Speed response under external disturbance in the experiments.

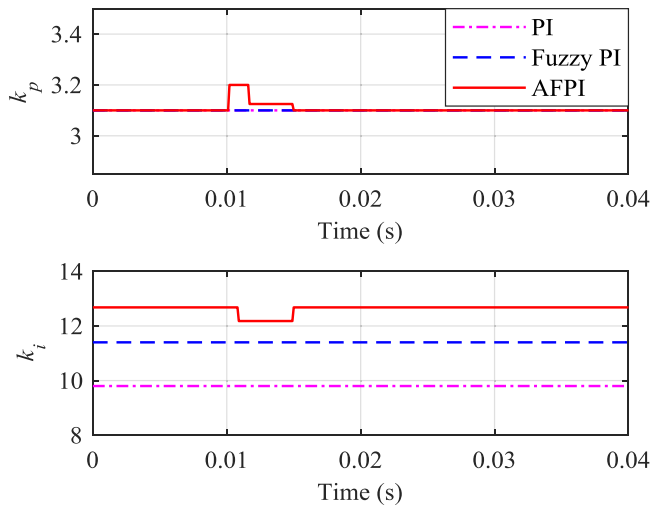


Fig. 24. Parameter (k_p , k_i) update curves for three controllers in the experiments.

illustrates the ability of the proposed AFPI control to deal with external disturbances such as nonlinear torque disturbances. The parameter updating curves and controller output curves are shown in Figs. 24 and 25, respectively. After the disturbance occurred suddenly, neither the fuzzy PI control nor the PI control adaptively adjusts the parameters. The fuzzy PI control system mainly relies on the optimal parameters determined at the startup stage to improve robustness. The proposed AFPI control method can further regulate the control parameters to the system under disturbances by an adaptive fuzzy tuner. This indicates the effective parameter tuning function of the AFPI control strategy strengthens the anti-disturbance capability while maintaining excellent transient response performance. In this context, the proposed scheme can further expand the potential of

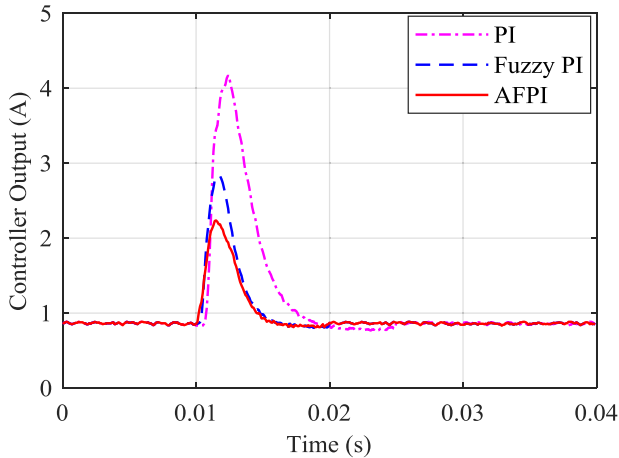


Fig. 25. Controller outputs in experiments related to external disturbance.

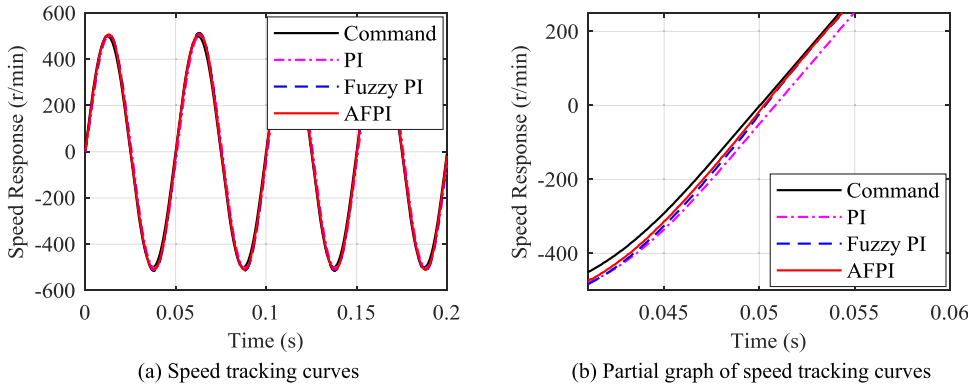


Fig. 26. System speed tracking response when using PI, fuzzy PI, AFPI in the experiments.

fuzzy PI control in dealing with the sudden change of the system state caused by the uncertainty of the inductance or resistance parameters.

Furthermore, the tracking performance of the PMSM drive system is tested under the three comparison controllers. The considered input speed command is the sinusoidal signal with amplitude being 500 r/min and a frequency being 20Hz. As shown in Fig. 26, the speed tracking curve of the AFPI control method is closer to the command curve than that of the other two methods. The results in Fig. 27 present PMSM system with the AFPI method obtains the smallest tracking error. As a further observation, Table 5 also gives speed tracking performance indexes. In terms of the root mean square error, the values of the PI method, the fuzzy PI, and the AFPI method are 36.1 r/min, 20.02 r/min, and 12.24 r/min, respectively. These results show the superior of the AFPI control in the tracking performance. As shown in Fig. 28, this is because the gain of the proposed AFPI control can be effectively adjusted adaptively with the change of the tracking error. And the controller output curves of the compared control system are plotted in Fig. 29. Due to the effective parameter tuning ability, the proposed AFPI

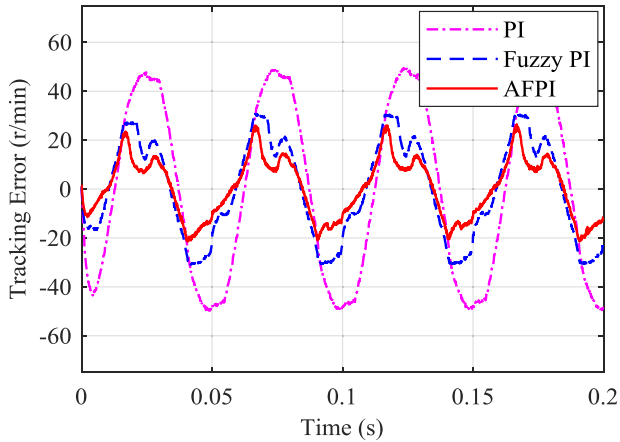


Fig. 27. Speed tracking error when using PI, fuzzy PI, AFPI in the experiments.

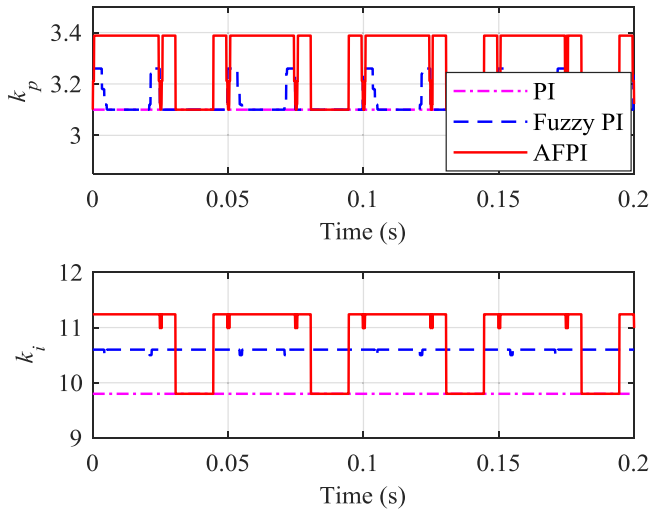


Fig. 28. Parameter (k_p , k_i) update curves for three controllers in the experiments.

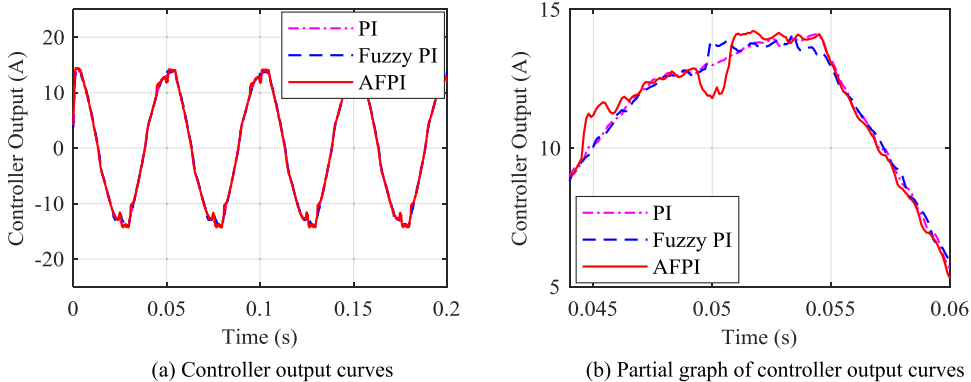


Fig. 29. Controller outputs in experiments related to speed tracking.

method can ensure that the controller output traces the speed error changes in time, thereby providing superior tracking performance for the PMSM speed control system.

6. Conclusion

This paper presented a novel AFPI control scheme for the PMSM drive system based on PFC. Specifically, conventional PFC was improved to predict future states. Then, through using the predicted output as the inputs, the parameters of the AFPI controller optimized in advance not only effectively dealt with the problem of system delay but also improved the robustness of the PMSM drive system. At the same time, the input domain of the fuzzy tuner was adjusted online to enhance the ability to deal with nonlinear uncertainties. The improved ASPI controller further mitigates the influence of input saturation on the control performance. Finally, the stability and the H_∞ robustness of the AFPI control system were analyzed based on the D-decomposition theory, which are guaranteed by the design criteria of the output scale factors. The method was applied to the simulation system and real-time PMSM drive system, and the results verified that the proposed strategy effectively improves the control accuracy and efficiency of the PMSM drive system as compared to traditional methods.

Declaration of Competing Interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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Erratum

Erratum to “Adaptive fuzzy PI controller for permanent magnet synchronous motor drive based on predictive functional control” [Journal of the Franklin Institute 360.15 (2023) 7333-7364]

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